Newman-Penrose Formalism Calculations

Stephani et al., Equation (16.46)

Metric

$$ds^2 = (\frac{h-1}{h+1})\mathrm{d}t \otimes \mathrm{d}t + (\frac{16\,(h^4+4\,h^3+6\,h^2+4\,h+1)e^g}{R^4r^4+8\,R^2r^2+16})\mathrm{d}r \otimes \mathrm{d}r + (\frac{16\,(r^2h^4+4\,r^2h^3+6\,r^2h^2+4\,r^2h+r^2)e^g}{R^4r^4+8\,R^2r^2+16})\mathrm{d}\theta \otimes \mathrm{d}\theta + (\frac{16\,(r^2h^4+4\,r^2h^3+6\,r^2h^2+4\,r^2h+r^2)e^g\sin(\theta)^2}{R^4r^4+8\,R^2r^2+16})\mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$(x^\mu) = (t, r, \theta, \phi)$

Metric Functions

$$h = \frac{1}{4}\sqrt{R^2r^2+4}mre^{(-\frac{1}{2}\,g)}$$

$$g(t) = g$$

Null Tetrad

Covariant components

$$l = (\sqrt{\frac{1}{2}}\sqrt{-\frac{h}{h+1}+\frac{1}{h+1}})\,dt + (-4\sqrt{\frac{1}{2}}\sqrt{\frac{h^4}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h^3}{R^4r^4+8\,R^2r^2+16}+\frac{6\,h^2}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h}{R^4r^4+8\,R^2r^2+16}+\frac{1}{R^4r^4+8\,R^2r^2+16}}e^{(\frac{1}{2}\,g)})\,dr$$

$$n = (\sqrt{\frac{1}{2}}\sqrt{-\frac{h}{h+1}+\frac{1}{h+1}})\,dt + (4\sqrt{\frac{1}{2}}\sqrt{\frac{h^4}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h^3}{R^4r^4+8\,R^2r^2+16}+\frac{6\,h^2}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h}{R^4r^4+8\,R^2r^2+16}+\frac{1}{R^4r^4+8\,R^2r^2+16}}e^{(\frac{1}{2}\,g)})\,dr$$

$$m = (4\sqrt{\frac{1}{2}}\sqrt{\frac{h^4}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h^3}{R^4r^4+8\,R^2r^2+16}+\frac{6\,h^2}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h}{R^4r^4+8\,R^2r^2+16}+\frac{1}{R^4r^4+8\,R^2r^2+16}}re^{(\frac{1}{2}\,g)})\,d\theta + (4i\sqrt{\frac{1}{2}}\sqrt{\frac{h^4}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h^3}{R^4r^4+8\,R^2r^2+16}+\frac{6\,h^2}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h}{R^4r^4+8\,R^2r^2+16}+\frac{1}{R^4r^4+8\,R^2r^2+16}}re^{(\frac{1}{2}\,g)}\sin(\theta))\,d\phi$$

$$\overline{m} = (4\sqrt{\frac{1}{2}}\sqrt{\frac{h^4}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h^3}{R^4r^4+8\,R^2r^2+16}+\frac{6\,h^2}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h}{R^4r^4+8\,R^2r^2+16}+\frac{1}{R^4r^4+8\,R^2r^2+16}}re^{(\frac{1}{2}\,g)})\,d\theta$$
$$+(-4i\sqrt{\frac{1}{2}}\sqrt{\frac{h^4}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h^3}{R^4r^4+8\,R^2r^2+16}+\frac{6\,h^2}{R^4r^4+8\,R^2r^2+16}+\frac{4\,h}{R^4r^4+8\,R^2r^2+16}+\frac{1}{R^4r^4+8\,R^2r^2+16}}re^{(\frac{1}{2}\,g)}\sin(\theta))\,d\phi$$

Contravariant components

$$l = (\frac{\sqrt{h+1}\sqrt{-h+1}}{\sqrt{2}h-\sqrt{2}})\,dt + (-\frac{R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}h^2+2\,\sqrt{2}h+\sqrt{2})} - \frac{e^{(-\frac{1}{2}\,g)}}{\sqrt{2}h^2+2\,\sqrt{2}h+\sqrt{2}})\,dr$$

$$n = (\frac{\sqrt{h+1}\sqrt{-h+1}}{\sqrt{2}h-\sqrt{2}})\,dt + (\frac{R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}h^2+2\,\sqrt{2}h+\sqrt{2})} + \frac{e^{(-\frac{1}{2}\,g)}}{\sqrt{2}h^2+2\,\sqrt{2}h+\sqrt{2}})\,dr$$

$$m = (\frac{R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)} + \frac{e^{(-\frac{1}{2}\,g)}}{\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r})\,d\theta + (\frac{i\,R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)\sin(\theta)} + \frac{i\,e^{(-\frac{1}{2}\,g)}}{(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)\sin(\theta)})\,d\phi$$

$$\overline{m} = (\frac{R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)} + \frac{e^{(-\frac{1}{2}\,g)}}{\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r})\,d\theta + (-\frac{i\,R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)\sin(\theta)} - \frac{i\,e^{(-\frac{1}{2}\,g)}}{(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)\sin(\theta)})\,d\phi$$

Directional Derivatives

$$D = \frac{\sqrt{h+1}\sqrt{-h+1}}{\sqrt{2}h-\sqrt{2}}\,\partial_t - \frac{R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}h^2+2\,\sqrt{2}h+\sqrt{2})} - \frac{e^{(-\frac{1}{2}\,g)}}{\sqrt{2}h^2+2\,\sqrt{2}h+\sqrt{2}}\,\partial_r$$

$$\Delta = \frac{\sqrt{h+1}\sqrt{-h+1}}{\sqrt{2}h-\sqrt{2}}\,\partial_t + \frac{R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}h^2+2\,\sqrt{2}h+\sqrt{2})} + \frac{e^{(-\frac{1}{2}\,g)}}{\sqrt{2}h^2+2\,\sqrt{2}h+\sqrt{2}}\,\partial_r$$

$$\delta = \frac{R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)} + \frac{e^{(-\frac{1}{2}\,g)}}{\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r}\,\partial_\theta + \frac{i\,R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)\sin(\theta)} + \frac{i\,e^{(-\frac{1}{2}\,g)}}{(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)\sin(\theta)}\,\partial_\phi$$

$$\overline{\delta} = \frac{R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)} + \frac{e^{(-\frac{1}{2}\,g)}}{\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r}\,\partial_\theta - \frac{i\,R^2r^2e^{(-\frac{1}{2}\,g)}}{4\,(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)\sin(\theta)} - \frac{i\,e^{(-\frac{1}{2}\,g)}}{(\sqrt{2}rh^2+2\,\sqrt{2}rh+\sqrt{2}r)\sin(\theta)}\,\partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}R^2r^3e^{(-\frac{1}{2}\,g)}h\frac{\partial}{\partial r}h}{4\,(rh^4+2\,rh^3-2\,rh-r)} - \frac{\sqrt{2}R^2r^2e^{(-\frac{1}{2}\,g)}h^2}{8\,(rh^4+2\,rh^3-2\,rh-r)} - \frac{\sqrt{2}R^2r^3e^{(-\frac{1}{2}\,g)}\frac{\partial}{\partial r}h}{4\,(rh^4+2\,rh^3-2\,rh-r)} - \frac{\sqrt{2}r\sqrt{-h+1}h^4\frac{\partial}{\partial t}g}{4\,(rh^4+2\,rh^3-2\,rh-r)\sqrt{h+1}} + \frac{\sqrt{2}R^2r^2e^{(-\frac{1}{2}\,g)}}{8\,(rh^4+2\,rh^3-2\,rh-r)}$$
$$- \frac{\sqrt{2}r\sqrt{-h+1}h^3\frac{\partial}{\partial t}g}{(rh^4+2\,rh^3-2\,rh-r)\sqrt{h+1}} - \frac{3\,\sqrt{2}r\sqrt{-h+1}h^2\frac{\partial}{\partial t}g}{2\,(rh^4+2\,rh^3-2\,rh-r)\sqrt{h+1}} + \frac{\sqrt{2}re^{(-\frac{1}{2}\,g)}h\frac{\partial}{\partial r}h}{rh^4+2\,rh^3-2\,rh-r} + \frac{\sqrt{2}e^{(-\frac{1}{2}\,g)}h^2}{2\,(rh^4+2\,rh^3-2\,rh-r)}$$
$$- \frac{\sqrt{2}r\sqrt{-h+1}h\frac{\partial}{\partial t}g}{(rh^4+2\,rh^3-2\,rh-r)\sqrt{h+1}} - \frac{\sqrt{2}re^{(-\frac{1}{2}\,g)}\frac{\partial}{\partial r}h}{rh^4+2\,rh^3-2\,rh-r} - \frac{\sqrt{2}r\sqrt{-h+1}\frac{\partial}{\partial t}g}{4\,(rh^4+2\,rh^3-2\,rh-r)\sqrt{h+1}} - \frac{\sqrt{2}e^{(-\frac{1}{2}\,g)}}{2\,(rh^4+2\,rh^3-2\,rh-r)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}R^2r^3e^{(-\frac{1}{2}\,g)}h\frac{\partial}{\partial r}h}{4\,(rh^4+2\,rh^3-2\,rh-r)} - \frac{\sqrt{2}R^2r^2e^{(-\frac{1}{2}\,g)}h^2}{8\,(rh^4+2\,rh^3-2\,rh-r)} - \frac{\sqrt{2}R^2r^3e^{(-\frac{1}{2}\,g)}\frac{\partial}{\partial r}h}{4\,(rh^4+2\,rh^3-2\,rh-r)} + \frac{\sqrt{2}r\sqrt{-h+1}h^4\frac{\partial}{\partial t}g}{4\,(rh^4+2\,rh^3-2\,rh-r)\sqrt{h+1}} + \frac{\sqrt{2}R^2r^2e^{(-\frac{1}{2}\,g)}}{8\,(rh^4+2\,rh^3-2\,rh-r)}$$
$$+ \frac{\sqrt{2}r\sqrt{-h+1}h^3\frac{\partial}{\partial t}g}{(rh^4+2\,rh^3-2\,rh-r)\sqrt{h+1}} + \frac{3\,\sqrt{2}r\sqrt{-h+1}h^2\frac{\partial}{\partial t}g}{2\,(rh^4+2\,rh^3-2\,rh-r)\sqrt{h+1}} + \frac{\sqrt{2}re^{(-\frac{1}{2}\,g)}h\frac{\partial}{\partial r}h}{rh^4+2\,rh^3-2\,rh-r} + \frac{\sqrt{2}e^{(-\frac{1}{2}\,g)}h^2}{2\,(rh^4+2\,rh^3-2\,rh-r)}$$
$$+ \frac{\sqrt{2}r\sqrt{-h+1}h\frac{\partial}{\partial t}g}{(rh^4+2\,rh^3-2\,rh-r)\sqrt{h+1}} - \frac{\sqrt{2}re^{(-\frac{1}{2}\,g)}\frac{\partial}{\partial r}h}{rh^4+2\,rh^3-2\,rh-r} + \frac{\sqrt{2}r\sqrt{-h+1}\frac{\partial}{\partial t}g}{4\,(rh^4+2\,rh^3-2\,rh-r)\sqrt{h+1}} - \frac{\sqrt{2}e^{(-\frac{1}{2}\,g)}}{2\,(rh^4+2\,rh^3-2\,rh-r)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}R^2r^2\cos(\theta)e^{(-\frac{1}{2}\,g)}}{16\,(rh^2+2\,rh+r)\sin(\theta)} - \frac{\sqrt{2}\cos(\theta)e^{(-\frac{1}{2}\,g)}}{4\,(rh^2+2\,rh+r)\sin(\theta)}$$

$$\beta = \frac{\sqrt{2}R^2r^2\cos(\theta)e^{(-\frac{1}{2}\,g)}}{16\,(rh^2+2\,rh+r)\sin(\theta)} + \frac{\sqrt{2}\cos(\theta)e^{(-\frac{1}{2}\,g)}}{4\,(rh^2+2\,rh+r)\sin(\theta)}$$

$$\tau = 0$$

$\pi = 0$

$$\epsilon = -\frac{R^2 r^2 e^{(-\frac{1}{2}\,g)} \frac{\partial}{\partial r} h}{8\,(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})} + \frac{\sqrt{-h+1}h^4 \frac{\partial}{\partial t} g}{4\,(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})\sqrt{h+1}} + \frac{\sqrt{-h+1}h^3 \frac{\partial}{\partial t} g}{(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})\sqrt{h+1}} + \frac{3\,\sqrt{-h+1}h^2 \frac{\partial}{\partial t} g}{2\,(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})\sqrt{h+1}} + \frac{\sqrt{-h+1}h \frac{\partial}{\partial t} g}{(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})\sqrt{h+1}} - \frac{e^{(-\frac{1}{2}\,g)} \frac{\partial}{\partial r} h}{2\,(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})} + \frac{\sqrt{-h+1} \frac{\partial}{\partial t} g}{4\,(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})\sqrt{h+1}}$$

$$\gamma = -\frac{R^2 r^2 e^{(-\frac{1}{2}\,g)} \frac{\partial}{\partial r} h}{8\,(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})} - \frac{\sqrt{-h+1}h^4 \frac{\partial}{\partial t} g}{4\,(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})\sqrt{h+1}} - \frac{\sqrt{-h+1}h^3 \frac{\partial}{\partial t} g}{(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})\sqrt{h+1}} - \frac{3\,\sqrt{-h+1}h^2 \frac{\partial}{\partial t} g}{2\,(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})\sqrt{h+1}} - \frac{\sqrt{-h+1}h \frac{\partial}{\partial t} g}{(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})\sqrt{h+1}} - \frac{e^{(-\frac{1}{2}\,g)} \frac{\partial}{\partial r} h}{2\,(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})} - \frac{\sqrt{-h+1} \frac{\partial}{\partial t} g}{4\,(\sqrt{2}h^4 + 2\,\sqrt{2}h^3 - 2\,\sqrt{2}h - \sqrt{2})\sqrt{h+1}}$$

Ricci Tensor Components

$\Phi_{01} = 0$

$\Phi_{02} = 0$

$\Phi_{10} = 0$

$\Phi_{12} = 0$

$\Phi_{20} = 0$

$\Phi_{21} = 0$

$$
\begin{aligned}
\Phi_{00} = {} & \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^5 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^4 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^4 r^5 e^{(-g)} h \frac{\partial}{\partial r} h^2}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^4 r^4 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{32\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^3 \sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h^3 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^4 r^5 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^4 r^4 e^{(-g)} h \frac{\partial}{\partial r} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{3\,R^2 r^3 \sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h^2 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^5 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{rh^8 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^4 e^{(-g)} \frac{\partial}{\partial r} h}{32\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{3\,R^2 r^3 \sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^3 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{3\,rh^7 \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^2 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^3 \sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^2 r^3 e^{(-g)} h \frac{\partial}{\partial r} h^2}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r e^{(-g)} h^4}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{7\,rh^6 \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^2 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{r\sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h^3 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{7\,rh^5 \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^2 e^{(-g)} h \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{3\,r\sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h^2 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^3 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r e^{(-g)} h^2}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& - \frac{R^2 r^2 e^{(-g)} \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{3\,r\sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{r e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& - \frac{r e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} - \frac{7\,rh^3 \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{e^{(-g)} h^3 \frac{\partial}{\partial r} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& - \frac{r\sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{r e^{(-g)} h \frac{\partial}{\partial r} h^2}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} + \frac{r e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& + \frac{R^2 r e^{(-g)}}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{7\,rh^2 \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{3\,e^{(-g)} h^2 \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{r e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} - \frac{3\,rh \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{e^{(-g)} h \frac{\partial}{\partial r} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& - \frac{r e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} - \frac{r \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{3\,e^{(-g)} \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)}
\end{aligned}
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\begin{aligned}
\Phi_{11} = & -\frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^4 r^4 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^4 r^5 e^{(-g)} h \frac{\partial}{\partial r} h^2}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{64\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^4 r^4 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{32\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^5 e^{(-g)} \frac{\partial}{\partial r} h^2}{64\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^4 r^4 e^{(-g)} h \frac{\partial}{\partial r} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^5 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{64\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{rh^8 \frac{\partial^2}{(\partial t)^2} g}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^4 r^4 e^{(-g)} \frac{\partial}{\partial r} h}{32\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{3\,rh^7 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h \frac{\partial}{\partial r} h^2}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r e^{(-g)} h^4}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{7\,rh^6 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^2 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^3 e^{(-g)} \frac{\partial}{\partial r} h^2}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{7\,rh^5 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^3 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r e^{(-g)} h^2}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^2 r^2 e^{(-g)} \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{r e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} - \frac{7\,rh^3 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{e^{(-g)} h^3 \frac{\partial}{\partial r} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} + \frac{r e^{(-g)} h \frac{\partial}{\partial r} h^2}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} + \frac{r e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^2 r e^{(-g)}}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{7\,rh^2 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{e^{(-g)} h^2 \frac{\partial}{\partial r} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& - \frac{r e^{(-g)} \frac{\partial}{\partial r} h^2}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{3\,rh \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{e^{(-g)} h \frac{\partial}{\partial r} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& - \frac{r e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{r \frac{\partial^2}{(\partial t)^2} g}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{e^{(-g)} \frac{\partial}{\partial r} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r}
\end{aligned}
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\begin{aligned}
\Phi_{22} = {} & \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^5 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^4 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^4 r^5 e^{(-g)} h \frac{\partial}{\partial r} h^2}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^4 r^4 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{32\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^3 \sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h^3 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^4 r^5 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^4 r^4 e^{(-g)} h \frac{\partial}{\partial r} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{3\,R^2 r^3 \sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h^2 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^5 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{rh^8 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^4 e^{(-g)} \frac{\partial}{\partial r} h}{32\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{3\,R^2 r^3 \sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^3 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{3\,rh^7 \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^2 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^3 \sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^2 r^3 e^{(-g)} h \frac{\partial}{\partial r} h^2}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r e^{(-g)} h^4}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{7\,rh^6 \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^2 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{r\sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h^3 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{7\,rh^5 \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^2 e^{(-g)} h \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{3\,r\sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h^2 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^3 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r e^{(-g)} h^2}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& - \frac{R^2 r^2 e^{(-g)} \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{3\,r\sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} h \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{r e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& - \frac{r e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} - \frac{7\,rh^3 \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{e^{(-g)} h^3 \frac{\partial}{\partial r} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& + \frac{r\sqrt{h+1}\sqrt{-h+1} e^{(-\frac{1}{2}\,g)} \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{r e^{(-g)} h \frac{\partial}{\partial r} h^2}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} + \frac{r e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& + \frac{R^2 r e^{(-g)}}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{7\,rh^2 \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{3\,e^{(-g)} h^2 \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{r e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} - \frac{3\,rh \frac{\partial^2}{(\partial t)^2} g}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{e^{(-g)} h \frac{\partial}{\partial r} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& - \frac{r e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} - \frac{r \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{3\,e^{(-g)} \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)}
\end{aligned}
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$$
\begin{aligned}
\Lambda = & -\frac{R^4 r^5 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{48\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{64\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^4 r^5 e^{(-g)} \frac{\partial}{\partial r} h^2}{192\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^4 r^5 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{48\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{rh^8 \frac{\partial}{\partial t} g^2}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^5 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{64\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{3\,rh^7 \frac{\partial}{\partial t} g^2}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{rh^8 \frac{\partial^2}{(\partial t)^2} g}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^3 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{6\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{7\,rh^6 \frac{\partial}{\partial t} g^2}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{3\,rh^7 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^2 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{6\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r e^{(-g)} h^4}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{7\,rh^5 \frac{\partial}{\partial t} g^2}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{7\,rh^6 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^2 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^3 e^{(-g)} \frac{\partial}{\partial r} h^2}{24\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{6\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{7\,rh^5 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^2 e^{(-g)} h \frac{\partial}{\partial r} h}{6\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^2 r^3 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r e^{(-g)} h^2}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{7\,rh^3 \frac{\partial}{\partial t} g^2}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^2 r^2 e^{(-g)} \frac{\partial}{\partial r} h}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{re^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{3\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{7\,rh^2 \frac{\partial}{\partial t} g^2}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{7\,rh^3 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{2\,e^{(-g)} h^3 \frac{\partial}{\partial r} h}{3\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{re^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^2 re^{(-g)}}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{3\,rh \frac{\partial}{\partial t} g^2}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{7\,rh^2 \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{e^{(-g)} h^2 \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{re^{(-g)} \frac{\partial}{\partial r} h^2}{12\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{re^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{3\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{r \frac{\partial}{\partial t} g^2}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{3\,rh \frac{\partial^2}{(\partial t)^2} g}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{2\,e^{(-g)} h \frac{\partial}{\partial r} h}{3\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{re^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{r \frac{\partial^2}{(\partial t)^2} g}{8\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{e^{(-g)} \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)}
\end{aligned}
$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_3 = 0$$

$\Psi_4 = 0$

$$
\begin{aligned}
\Psi_2 = {} & \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^4 r^5 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{48\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^4 r^4 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{3\,R^4 r^5 e^{(-g)} h \frac{\partial}{\partial r} h^2}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{32\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{3\,R^4 r^4 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{32\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{11\,R^4 r^5 e^{(-g)} \frac{\partial}{\partial r} h^2}{96\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^4 r^5 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{48\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^4 r^4 e^{(-g)} h \frac{\partial}{\partial r} h}{16\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^4 r^5 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{32\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{3\,R^4 r^4 e^{(-g)} \frac{\partial}{\partial r} h}{32\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^3 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{6\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^2 r^2 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{6\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{3\,R^2 r^3 e^{(-g)} h \frac{\partial}{\partial r} h^2}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^2 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{11\,R^2 r^3 e^{(-g)} \frac{\partial}{\partial r} h^2}{12\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{R^2 r^3 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{6\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{R^2 r^2 e^{(-g)} h \frac{\partial}{\partial r} h}{6\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{R^2 r^3 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{R^2 r^2 e^{(-g)} \frac{\partial}{\partial r} h}{4\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{r e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} \\
& - \frac{r e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{3\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{e^{(-g)} h^3 \frac{\partial}{\partial r} h}{3\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{3\,r e^{(-g)} h \frac{\partial}{\partial r} h^2}{rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r} + \frac{r e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{e^{(-g)} h^2 \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{11\,r e^{(-g)} \frac{\partial}{\partial r} h^2}{6\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& + \frac{r e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{3\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} - \frac{e^{(-g)} h \frac{\partial}{\partial r} h}{3\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} \\
& - \frac{r e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)} + \frac{e^{(-g)} \frac{\partial}{\partial r} h}{2\,(rh^8 + 4\,rh^7 + 4\,rh^6 - 4\,rh^5 - 10\,rh^4 - 4\,rh^3 + 4\,rh^2 + 4\,rh + r)}
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"